



Is AI a 'substance driver' or a 'fiction booster'? The impact of AI application on corporate green innovation bubbles

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ABSTRACT

This study investigates whether artificial intelligence (AI) acts as a "substance driver" that promotes genuine green innovation or a "fiction booster" that facilitates sophisticated greenwashing. Analyzing panel data from Chinese A-share listed manufacturing companies spanning 2013 to 2023, fixed effects regression results reveal that AI application significantly suppresses green innovation bubbles, demonstrating a substantial reduction effect at the sample mean. However, this effect is highly contingent on external institutional and competitive contexts. Stringent environmental regulation amplifies AI's bubble-suppressing effect by channeling AI capabilities toward substantive compliance-oriented innovation. Additionally, intense market competition amplifies this effect, as competitive pressure disciplines firms to deploy AI toward substantive innovations that generate genuine competitive advantages rather than superficial green credentials. Heterogeneity analyses reveal AI's effectiveness is significantly stronger among non-state-owned enterprises and non-heavy-polluting firms, while SOEs and heavy-polluting firms show insignificant effects. Firms located in AI innovation pilot zones exhibit stronger bubble-suppressing effects compared to those outside. These findings contribute an integrated "institution-market" contingency framework to technology adoption literature, introduce an objective patent-based measure of innovation quality, and demonstrate that AI's role in corporate sustainability is neither technologically deterministic nor universally beneficial but critically depends on the alignment of institutional pressures and market incentives. The results remain robust to instrumental variable analysis, propensity score matching, Heckman correction, and multiple robustness checks.

1. Introduction

1.1. The paradox of green innovation in the era of carbon neutrality

The global imperative to address climate change has placed corporate green innovation at the forefront of both policy agendas and strategic management discourse (Amore and Bannedsen, 2016; Liu et al., 2024; Tian et al., 2024; Dai et al., 2025a). As nations worldwide commit to ambitious decarbonization targets, such as China's pledge to achieve carbon peak by 2030 and carbon neutrality by 2060, governments have deployed an unprecedented array of policy instruments to stimulate corporate investment in

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